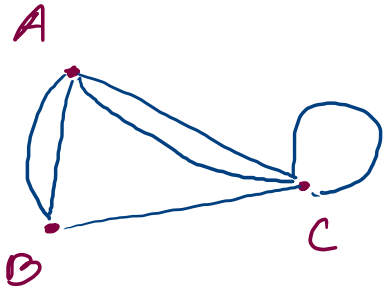


10.1 Graphs & features

⑩ Defⁿ. 1

$G(V, E)$
↓
set of edges.
↓
set of vertices

vertices $\begin{cases} \rightarrow \text{finite} \rightarrow \text{finite graphs.} \\ \rightarrow \text{infinite} \rightarrow \text{infinite "} \end{cases}$

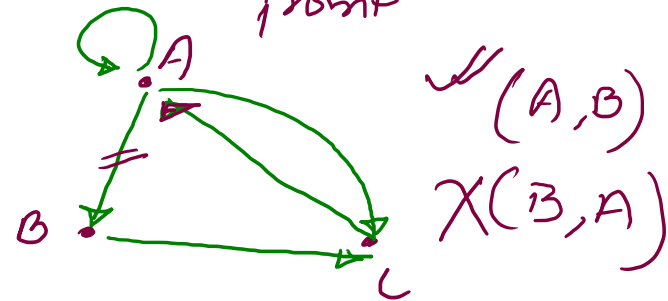


undirected graphs.
(UG)

⑩ Defⁿ 2 Directed graphs.
Di-graphs
(DG)

$G(V, E)$
↓
set of vertices
elements in this set
are ordered pairs

$e_i = (u_i, v_i)$
↓
initial point → terminal node.

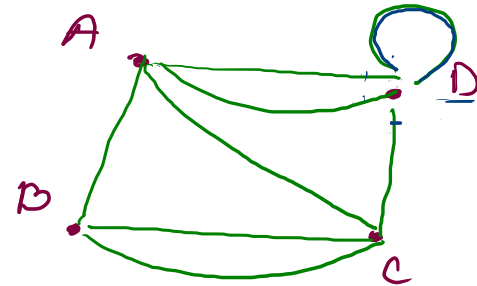


① vertex of degree zero $\rightarrow$ isolated vertex $\rightarrow \bullet$
one $\rightarrow$ pendant $\rightarrow \bullet - \text{---}$

② Theorem - 1 hand-shaking theorem.

UG $\rightarrow G(V, E) \rightarrow m$ edges.

$$2m = \sum_{v \in V} \deg(v)$$



$$2m = 8 \times 2 = 16$$

$$\deg(A) = 4$$

$$\deg(B) = 3$$

$$\deg(C) = 4$$

$$\deg(D) = 5$$

$$\sum_{v \in V} \deg(v) = 16$$

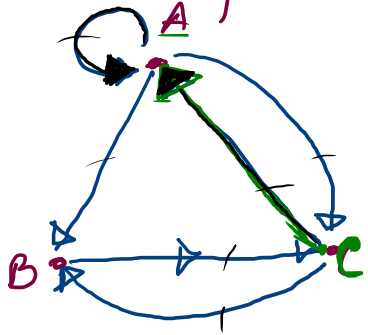
③ Theorem - 2 An UG has an even number of vertices with odd degrees.

$$\rightarrow 2m = \sum_{v \in V} \deg(v) \quad V \begin{cases} V_1 \rightarrow \text{set of even degree vertices} \\ V_2 \rightarrow \text{odd vertices} \end{cases}$$

$$\Rightarrow 2m = \underbrace{\sum_{v_1 \in V_1} \deg(v_1)}_{\text{even}} + \underbrace{\sum_{v_2 \in V_2} \deg(v_2)}_{|V_2| = \text{even}} \rightarrow \text{proven}$$

$$\begin{aligned} 5 + 3 &\rightarrow 8 \\ 4 + 1 &\rightarrow 5 \\ 4 + 1 + 3 &= 8 \end{aligned}$$

(*) Defⁿ 5



$V_i \in V_{DG}$ $\left\{ \begin{array}{l} \text{in-degree} \rightarrow v_i \text{ as terminal} \rightarrow \deg^-(v_i) \\ \text{vertex} \\ \text{out-degree} \rightarrow v_i \text{ as initial} \rightarrow \deg^+(v_i) \\ \text{vertex} \end{array} \right.$

v	$\deg^-(v)$	$\deg^+(v)$
A	2	3
B	2	1
C	2	2
Σ	6	6

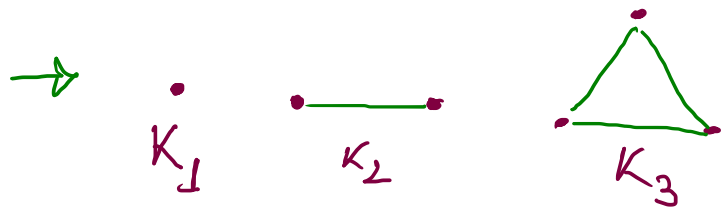
(*) Theorem-3 In a DG $G=(V,E)$ we have

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$$

Types of graphs.

① Complete graphs (K_n)

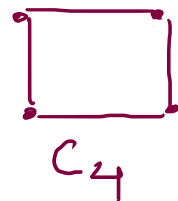
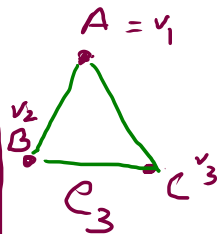
A graph with exactly one edge between each pair of distinct vertices.



② Cycles (C_n), $n \geq 3$

$$C_n = (V, E)$$

$$E = \{ \{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\} \}$$

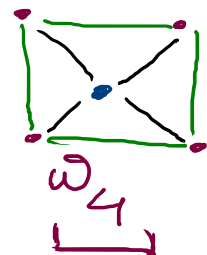


③ Wheels.

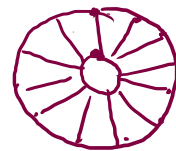
$$W_n = \underline{C_n} + \underline{1}$$



$$+ \underline{1} =$$



$$W_n = W_{|V|-1}$$

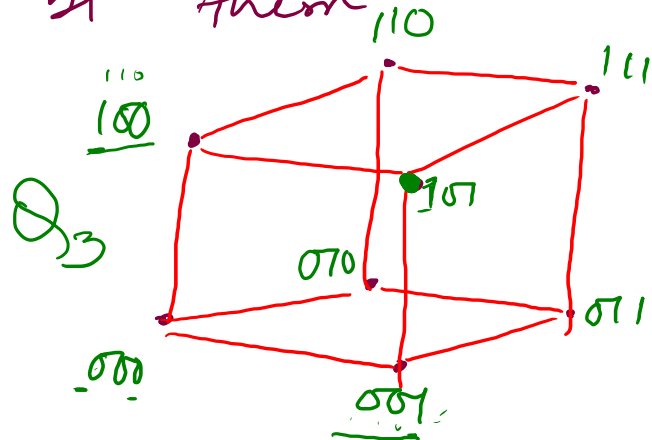


④ n-cubes $\rightarrow \mathcal{Q}_n$

- $2^n \rightarrow$ vertices $\rightarrow$ n-dimensional hypercubes.
 $|n| \rightarrow$ bitstring.

$\mathcal{Q}_3 \rightarrow$ 8 vertices
 $\rightarrow$ Cube

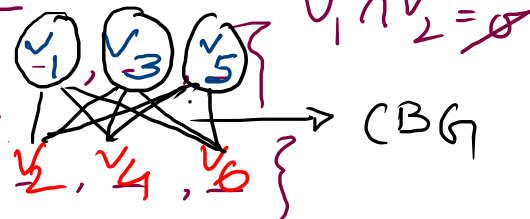
- All vertices that are adjacent will have exactly 1 bit difference b/w them



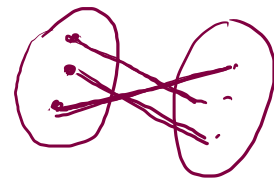
⑤ Bipartite graphs

$G(V, E)$

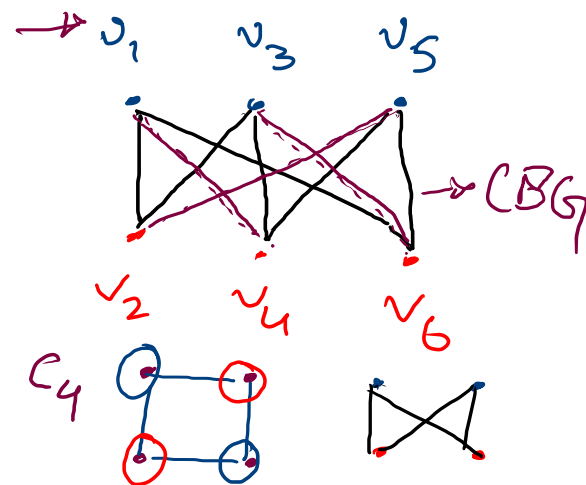
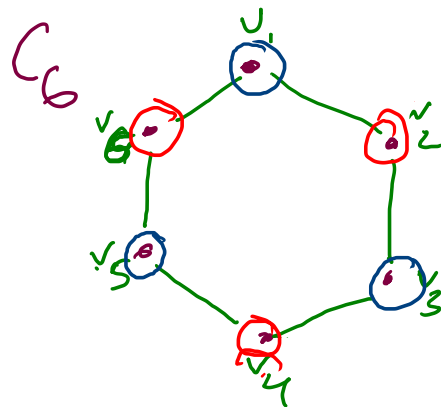
$V \leftarrow \begin{matrix} v_1 \\ v_2 \end{matrix}$



* A graph where V can be divided into 2 sets (disjoint) v_1, v_2

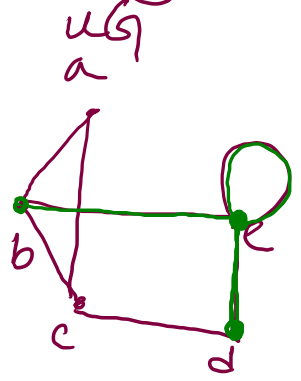


such that every edge in either v_1 or v_2 are not adjacent.

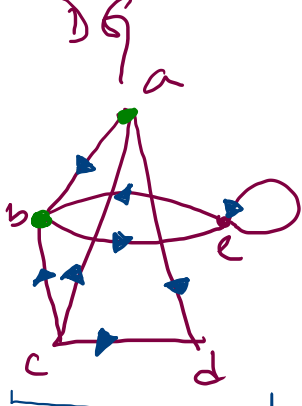


10.3 Graph representation

① Adjacency list

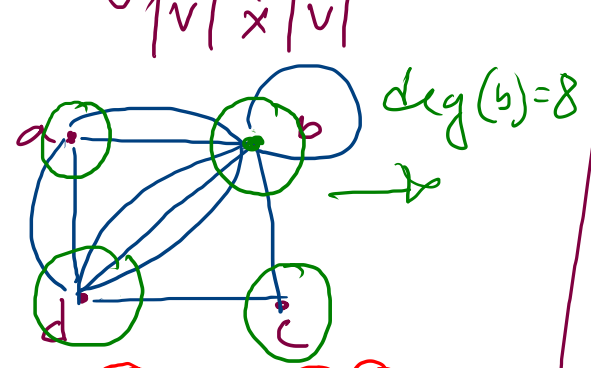


Vertices	Adj _{uv}
a	b, c
b	c, a, e
c	a, b, d
d	c, e
e	d, b, e



V_i	V_f
a	b, d
b	e
c	b, a, d
d	-
e	e, b

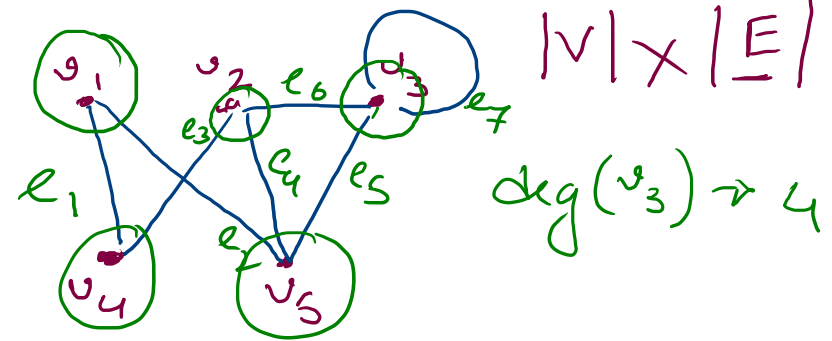
② Adjacency Matrix



	a	b	c	d	
a	0	2	0	2	→ 4
b	2	1	1	3	→ 7 +1
c	0	1	0	1	→ 2
d	2	3	1	0	→ 6

M_{ij} if $i=j$ and $M_{ij} > 0$ → loop

③ Incidence matrix



	e_1	e_2	e_3	e_4	e_5	e_6	e_7	
v_1	1	1	0	0	0	0	0	→ 2
v_2	0	0	1	1	0	1	0	→ 3
v_3	0	0	0	0	1	1	1	→ 3 +1
v_4	1	0	1	0	0	0	0	→ 2
v_5	0	1	0	1	1	0	0	→ 3